

PAPER_815: VDF vs GSMF SMBH Mass Function Proxy + $M_\bullet$ - σ Relation UQFF

Unified Quantum Field Framework — Whitepaper 815

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Abstract

This paper investigates two competing proxies for the SMBH mass function — the Galaxy Stellar Mass Function (GSMF) and the Velocity Dispersion Function (VDF) — within the Quadriadic UQFF framework. The VDF-derived gravitational wave background (GWB) amplitude $\log_{10} A_{yr} \approx -14.74$ differs from the GSMF-derived -14.9 , suggesting that the fundamental scaling relation is the $M_\bullet$ - σ relation rather than the $M_\bullet$ - M_* relation. The velocity dispersion σ_∞ is derived from the Sersic virial theorem and enters the UQFF Quadriadic Layer 3 (Buoyancy) as a gravitational velocity proxy.

1. Introduction

The NANOGrav GWB depends critically on the SMBH mass function, which in turn depends on whether we scale $M_\bullet$ via galaxy stellar mass M_* (GSMF approach) or stellar velocity dispersion σ (VDF approach). Observationally, massive compact galaxies (relic galaxies NGC 1271, NGC 1277) and LEGA-C survey data at $z = 0.6$ - 1.0 favor the VDF approach.

2. Characteristic Strain — GWB

The GWB characteristic strain:

$$h_c(f) = A_{yr} \cdot \left(\frac{f}{f_{yr}} \right)^{-2/3}$$

with individual binary strain:

$$h_s = \sqrt{\frac{32}{5}} \cdot \frac{(G\mathcal{M}_c/c^3)^{5/3} \cdot (\pi f_r)^{2/3}}{D_c}$$

where D_c = comoving distance, f_r = rest-frame orbital frequency.

3. VDF Number Density

The VDF-based SMBH binary merger number density:

$$\frac{d^3n}{dz dM_* dq} = \Phi_{VDF}(z, \sigma) \cdot R(z, M_*, q)$$

where Φ_{VDF} is the velocity dispersion function and R is the merger rate kernel. Integrating over the binary population:

$$A_{yr, VDF} \approx 10^{-14.74}$$

vs GSMF result:

$$A_{yr, GSMF} \approx 10^{-14.9}$$

The VDF yields a higher amplitude by ≈ 0.16 dex.

4. Velocity Dispersion Proxy

The effective velocity dispersion at the SMBH influence radius:

$$\sigma_\infty = \sqrt{\frac{GM_* K_*(n)}{r_e}}$$

where $K_*(n)$ is the Sersic virial constant (depends on Sersic index n), and r_e is the effective (half-light) radius. For typical early-type galaxies:

$$K_*(n) \approx \frac{73.32}{10.465 + (n - 0.94)^2} + 0.954$$

5. Orbital Evolution — Frequency Drift

The gravitational wave frequency evolution:

$$\frac{df_{orb}}{dt} = \frac{96}{5} \cdot \left(\frac{G\mathcal{M}_c}{c^3} \right)^{5/3} \cdot (2\pi)^{8/3} \cdot f_{orb}^{11/3}$$

This determines the binary's lifetime before merger from an initial separation a_0 .

6. $M_\bullet$ - σ Relation in Quadriadic UQFF

The fundamental $M_\bullet$ - σ scaling:

$$\log_{10} \left(\frac{M_\bullet}{10^9 M_\odot} \right) = \alpha_{M\sigma} \cdot \log_{10} \left(\frac{\sigma}{200 \text{ km/s}} \right) + \beta_{M\sigma}$$

with $\alpha_{M\sigma} \approx 4.4$, $\beta_{M\sigma} \approx 0.3$. This enters UQFF Buoyancy Layer 3:

$$g_{L3,VDF} = \sigma_{\infty} \cdot \left(\frac{G}{r_e} \right)^{1/2} + \frac{GM_{\bullet}}{r_{inf}^2}$$

7. UQFF Buoyancy Layer 3 Integration

Full Buoyancy UQFF with VDF proxy:

$$g_{L3} = F_{U,Bi} + U_{i,buoyancy} + \sigma_{\infty} \cdot (G/r_e)^{1/2} + R_{merge} \cdot (GM_c/c^2)^{5/3}$$

The VDF contribution distinguishes high- σ compact galaxies (relic systems) from normal ellipticals through the $K_*(n)$ Sersic dependence.

8. Relic Galaxy Constraints

LEGA-C survey ($z = 0.6-1.0$) + local relic galaxies NGC 1271 ($M_* \approx 2 \times 10^{11} M_{\odot}$, $r_e \approx 2$ kpc) and NGC 1277 ($M_{\bullet}/M_{bulge} \approx 0.59$) verify that:

$$\sigma_{\infty}(\text{relic}) > \sigma_{\infty}(\text{normal elliptical})$$

at fixed M_* , confirming VDF captures more SMBH-encoded information than GSMF.

9. Summary

The VDF yields $A_{yr} \approx 10^{-14.74}$ for the GWB, consistent with NANOGrav-15yr measurements. The $M_{\bullet}-\sigma$ relation, through σ_{∞} , provides a superior proxy for SMBH mass and enters the Quadriadic UQFF Buoyancy Layer as a kinematic velocity correction.

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